

Lyu Zizheng

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STRENGTH SUMMARY

Results-oriented Senior Product & Systems Engineer with 5+ years of experience architecting high-concurrency distributed systems, financial payment engines, and production AI evaluation infrastructure. Tech Owner at Wise automating 30,000+ monthly payment defects (£80k/mo direct savings) via deterministic DAG state machines and shadowgating. Former ByteDance engineer (promoted in 1 year, Spot Bonus, Top E rating) mitigating multi-million QPS broadcast storms on TikTok IM. Pragmatic builder committed to business ROI, distributed consistency, and end-to-end product ownership.

EDUCATION

B.Eng. (Honors) in Computer Engineering

SG | Aug 2017 – May 2021

NANYANG TECHNOLOGICAL UNIVERSITY (NTU)

Scholarship: Full NTU Science and Engineering Undergraduate Merit Scholarship (Awarded to top 1%)

WORK EXPERIENCE

WISE | PRODUCT ENGINEER 3 (TECH OWNER)

SG | Nov 2024 – Present

- **Financial AI Automation & Evaluation Infrastructure:** Architected an end-to-end AI evaluation and execution platform from scratch for last-mile payment linking without golden datasets; engineered deterministic DAG state machines with real-time live shadowing against PayOps, 99.5% accuracy gating, and a 5% canary control stream with sub-second circuit-breaker failover.
- **Payment Defects & Direct Business Impact:** Automated 30,000+ complex payment defect and unallocated funds cases monthly with 98%+ accuracy across heterogeneous currency and clearing rail rules; designed risk-tiered blast radius controls (large-amount threshold gating to human approval), yielding **£80,000/month (~£1M/year)** direct operational savings.
- **Global Compensation Platform & Cross-Team Escalation:** Re-architected global multi-currency Compensation Platform with automated compliance checks and maker-checker controls; built a Kafka-based cross-team escalation pipeline between Customer Support and PayOps, leveraging structured defect telemetry to eliminate upstream ticket causes.

MARIBANK (SEA GROUP) | SENIOR BACKEND ENGINEER

SG | Apr 2024 – Jul 2024

- **Digital Banking Credit Core:** Engineered core credit services (Cashloan & SME Termloan) in Java, enforcing distributed transaction consistency, idempotency, and anti-replay patterns under MAS regulatory compliance; built an internal RAG-based LLM assistant for knowledge retrieval.

BONDEE | SENIOR SOFTWARE ENGINEER (EMPLOYEE #1 IN SG)

SG | Sep 2023 – Apr 2024

- **Cloud-Native Infrastructure:** Established foundational cloud-native infrastructure as employee #1 in Singapore; orchestrated multi-tenant K8s environments and deployed self-hosted Vector + Kafka + Elasticsearch observability pipelines to replace costly managed logging stacks.

BYTEDANCE | SENIOR BACKEND ENGINEER (PROMOTED IN 1 YEAR)

SG | Jan 2021 – Sep 2023

- **TikTok-Social-Instant Messaging (IM)**
 - **Presence Engine & Fan-out Storm Mitigation:** Engineered a hybrid push-pull protocol with active heart-beat gating via Redis ZSET to mitigate multi-million QPS broadcast storms during user status changes, eliminating $O(N \times M)$ write amplification by restricting push dispatches exclusively to online, active friends.
 - **Typing Indicator & Read Receipts:** Implemented ultra-low latency signaling mechanisms for Read Receipts and Typing Indicators across multi-data-center infrastructure; designed client-side debouncing and batch aggregation to eliminate transient signaling spikes while maintaining sub-100ms response.

- **End-to-End Latency Tuning & Cold Start Ranking:** Overhauled inbox cold-start loading via ZSET active conversation hierarchies to replace append-only DB table scans (delivering sub-100ms initialization for newly re-installed apps); optimized cross-region RPC payloads and routing paths to reduce P99 latency across core messaging services.
- **Domain Ownership & Security Compliance:** Served as Tech Owner for IM core microservices; mentored new engineers and led cross-functional safety/security technical initiatives ensuring compliance with US security standards and EU GDPR.
- **Location System Cloud Platform** (IP Geo-Location PoC | 2022 Jan–Feb Spot Bonus Receiver | Top E Rating)
 - **Overseas Location Governance & Optimization:** Sole engineer in Singapore independently governing overseas IP geo-location infrastructure (~100 offline Spark batch jobs and Go microservices); overhauled alert rules to eliminate oncall alarm fatigue, tuned Go runtime memory/GC profiles, and significantly improved ASEAN geo-accuracy, earning a **Spot Bonus** and top **Exceeds Expectations (E)** rating.
- **API Gateway (AGW) Cloud Platform** (Intern)
 - Developed automated self-diagnosis and capacity estimation tools for PaaS gateway routing millions of QPS during Chinese New Year (CNY) live traffic campaigns.

OTHER WORK EXPERIENCE

INDEPENDENT PRODUCTS | CREATOR & FULL-STACK BUILDER

SG | Aug 2024 – Present

- **CanCan** (<https://cancan.money>): Financial evidence ledger and automated reconciliation platform for independent businesses, enforcing double-entry bookkeeping principles.
- **CoffeeMode:** Cafe Map for digital nomads to find best places for remote working.
- **Our Village** (<https://gen-growth.com>): Decentralized community collaboration and membership management platform.

U-WAVE | CO-FOUNDER, FULL STACK ENGINEER

SG | Aug 2019 – Jan 2021

- Co-founded and built campus lifestyle app from scratch, scaling to 20,000 registered students and 4,000 DAU; engineered cross-platform mobile client with Flutter and architected backend microservices with Spring Cloud, Docker, and Redis.

SKILLS

Programming Languages: Golang, Java, TypeScript, JavaScript, Python, Dart, SQL, Shell

Distributed Systems: High-Concurrency Architecture, Microservices (gRPC/Thrift), Kafka, Redis, Distributed Consistency

Cloud & Infrastructure: Docker, Kubernetes, AWS, Vector, Elasticsearch, Apache Spark, CI/CD Pipelines, Linux

Domain Expertise: Financial Payment Core & Defect Automation, IM Protocols & Presence Engines, AI Evaluation Pipelines